

KARTHIKEYAN P

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience across three MERN stack internships and multiple independent and hackathon projects. Skilled in building secure, role-based, real-time web applications using React, Node.js, Express.js, and MongoDB, with working exposure to REST API design and geospatial data. Currently developing an AI-integrated hyperlocal donation platform incorporating computer vision, OCR, and route optimization. Strong collaborator with a research background in applied software engineering.

SKILLS

- Languages:** Python, Java, C, JavaScript (ES6+)
- Frontend:** React.js, React Native (CLI), HTML5, CSS3, Responsive Web Design
- Backend & APIs:** Node.js, Express.js, FastAPI, REST API Design, JWT Authentication
- Databases:** MongoDB, MongoDB Atlas (Geospatial Queries), MySQL
- AI / Computer Vision:** YOLOv8/YOLOv11, OpenCV, SAHI, OCR, Google Gemini API, MongoDB Vector Search & RAG
- Cloud Platforms:** Microsoft Azure Fundamentals, AWS (Security & Serverless Containers)
- Tools & Platforms:** Git, GitHub, Gradio, ReportLab, OSRM

CERTIFICATIONS

- Cloud:** Azure Fundamentals — CloudThat | Microsoft | NASSCOM
- AI & Data (MongoDB):** AI & Innovation: Resilient AI Strategy; Vector Search Fundamentals (Building AI-Powered Search); Retrieval-Augmented Generation (RAG) with MongoDB; Building AI Agents with MongoDB
- Security & Cloud (AWS):** AWS Security – Encryption Fundamentals; Choosing Serverless Containers for .NET — Infosys Springboard | AWS
- Databases (IIT Bombay):** RDBMS PostgreSQL; PHP and MySQL — Spoken Tutorial, IIT Bombay
- Academic (NPTEL, IIT):** Privacy & Security in Online Social Media; Cloud Computing; Ethical Hacking
- Bootcamps & Training:** MERN Stack Certification (Vinsup Technology); Front-End Development (AWLRI, Red Rivers Labs); Generative AI Bootcamp (Raam Techlink | NoviTech R&D)

WORK EXPERIENCE

MERN Stack Development Intern

Dec 2024 – Jan 2025

Vinsup Technology Pvt. Ltd.

Madurai, India

- Built backend web modules using MongoDB, Express.js, and Node.js to support scalable, REST API-driven applications.
- Developed responsive frontend interfaces with HTML, CSS, and JavaScript, ensuring consistent behavior across devices.
- Designed, tested, and debugged REST API endpoints under industry mentor supervision, improving reliability of data exchange between frontend and backend.

Web Developer, K.L.N. Innovation & Research Park (KLN IRP)

Dec 2024 – Present

K.L.N. College of Engineering

Madurai, India

- Working full-time as the web developer on an in-house research product, owning end-to-end development within the college innovation ecosystem.
- Collaborating closely with faculty mentors and a cross-functional team to build and refine the platform toward commercial readiness.
- Designing and testing end-to-end application workflows, improving execution speed, code quality, and overall product stability.

MERN Stack Development Intern

Jun 2026

Elysium Technologies Private Limited

Madurai, India

- Completed a selective, intensive MERN Stack engineering internship, building hands-on proficiency in React.js, Node.js, Express.js, and MongoDB through applied coding exercises.
- Developed production-style full-stack components following industry coding standards and version-controlled workflows.

PROJECTS

AI Resume PDF Generator

Aug 2025

Independent Lead Developer

- Built an end-to-end automated resume creation tool with a **Gradio-based web interface**, converting unstructured user input into structured resume content.

- Integrated the **Google Gemini API** to dynamically parse and restructure raw user input into targeted, role-specific resume text.
- Engineered a background PDF rendering pipeline using **ReportLab**, enabling instant, on-demand resume downloads.

E-MBook — Digital Measurement Book System

Jun 2026

Full Stack Developer (MERN) — Nimirndhu Nil Hackathon '26

- Developed a full-stack **MERN** application digitizing the manual Measurement Book process for the Rural Development & Panchayat Raj Department, replacing a paper-based workflow.
- Implemented role-based authentication and a **6-tier approval workflow** (JE → AE → AEE → EE → SE → CE) for hierarchical engineering approvals.
- Built integrated modules for project management, measurement entry, tender allocation, and automated reporting, connecting **MongoDB to Express.js REST APIs** for real-time monitoring dashboards.

Hospireo — Healthcare Discovery Platform

Testing Phase

Full Stack Engineer

- Built a hospital and doctor discovery platform with secure **Node.js/Express.js** backend APIs supporting real-time patient booking.
- Implemented **MongoDB** query logic for department-wise doctor profiles with real-time availability mapping.
- Currently in **testing phase**, with deployment planned for the near term.

HSDE — Hyperlocal Smart Donation Engine

In Progress — 45%

Full Stack & AI Systems Developer (Independent / IEEE Final-Year Project)

- Building a **mobile-centric** donation platform in **React Native (CLI)** connecting NGOs, donors, and volunteers across **10 donation categories** (food, medicines, medical equipment, blood, and more).
- Architecting a custom **Dynamic Donation Distribution Engine (DDSE)** to match NGO requests with donors within a **15 km radius**, prioritized by urgency, category, and location.
- Integrating **YOLO + OCR** for category-aware AI verification and **OSRM** for optimized volunteer routing, built on FastAPI and MongoDB Atlas.

EDUCATION

B.E., Computer Science and Engineering

Expected May 2027

K.L.N. College of Engineering — CGPA: 8.68 / 10 (through 6th Semester)

Higher Secondary Course (HSC)

2023

A.P.T. Dorairaj Higher Secondary School — 80.83%

ACHIEVEMENTS & LEADERSHIP

- Selected for Level 2 Ideation Camp, Nimirndhu Nil Hackathon '26 (EDII TN).
- Certified College-Level Participant, Smart India Hackathon.
- Presenter, KLNICST '26 International Conference.
- Participant, Pudhu Madurai Green MFG X Conference (CII – Madurai Zone).
- Technical Lead, Technozare Symposium.
- Technical Core Team Member, GDSC (Google Developer Student Clubs).
- Technical Team Support Member, Techgenio '26.
- Outreach Team Lead, CSE Symposium Zeigen '26.